

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

Let's break down the problem step by step to determine how far John is from home at the end of 4 hours .

1. Initial Drive to Realization :

- John drives for 3 hours at a speed of 60 mph .
- To find the distance covered in this initial drive , we use the formula :

$$\text{Distance} = \text{Speed} \times \text{Time}$$
$$= 60 \text{ mph} \times 3 \text{ hours} = 180 \text{ miles}$$

- Substituting the given values :

$$\text{Distance} = 60 \text{ mph} \times 3 \text{ hours} = 180 \text{ miles}$$

2. Return Trip :

- John turns around and starts his return trip home .
- He aims to get home in 4 hours .
- However , he spends the first 2 hours in stand still traffic , covering no distance :

$$\text{Distance} = 0 \text{ mph} \times 2 \text{ hours} = 0 \text{ miles}$$

3. Driving in Stand still Traffic :

- After the first 2 hours , he drives for the next half-hour (0.5 hours) at a speed of 30 mph .
- To find the distance covered in this segment :

$$\text{Distance} = 30 \text{ mph} \times 0.5 \text{ hours} = 15 \text{ miles}$$

4. Remaining Time :

- John has 4 hours total for the return trip .
- He has already used 2 hours in stand still traffic and 0.5 hours driving at 30 mph .
- The remaining time is :

$$4 \text{ hours} - 2 \text{ hours} - 0.5 \text{ hours} = 1.5 \text{ hours}$$

- During this remaining time , he drives at a speed of 80 mph .
- To find the distance covered in this segment :

$$\text{Distance} = 80 \text{ mph} \times 1.5 \text{ hours} = 120 \text{ miles}$$

5. Total Distance Covered :

- To find the total distance John has covered in the 4 hours , we sum up all the distances :

$$\text{Total Distance} = 0 \text{ miles} + 15 \text{ miles} + 120 \text{ miles} = 135 \text{ miles}$$

6. Distance from Home :

- John initially drove 180 miles away from home .
- He then returned a total of 135 miles .
- To find how far he is from home , we subtract the distance he has returned from the initial distance :

$$\text{Distance from Home} = 180 \text{ miles} - 135 \text{ miles} = 45 \text{ miles}$$

Thus , the distance from home at the end of the 4 hours is :

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1. Initial drive : 180 miles
2. Stand still traffic : 0 miles
3. Half-hour drive at 30 mph : 15 miles
4. Remaining time drive at 80 mph : 120 miles
5. Total return distance : 135 miles
6. Distance from home : 45 miles